

COMPONENTS:	ORIGINAL MEASUREMENTS:			
(1) 1-Butanol; C ₄ H ₁₀ O; [71-36-3]	Butler, J.A.V.; Thomson, D.W.; Macleman W.H.			
(2) Water; H ₂ O; [7732-18-5]	J. Chem. Soc. <u>1933</u> , 674-86.			
VARIABLES:	PREPARED BY:			
Temperature: 18-31°C	S.H. Yalkowsky; S.C. Valvani; A.F.M. Barton			
EXPERIMENTAL VALUES:				
Mutual solubility of 1-butanol(1) and water(2)				
t/°C	Alcohol-rich phase		Water-rich phase	
	(1)/100g sln	x ₁ (compiler)	g(1)/100g sln	x ₁ (compiler)
18.45	80.01	0.493	-	-
22.60	-	-	7.497	0.0193
23.40	79.73	0.489	-	-
23.70	-	-	7.407	0.0191
24.85	-	-	7.318	0.0188
25.00	79.64	0.488	7.31	0.0188
26.40	-	-	7.202	0.0185
27.45	79.50	0.486	-	-
28.06	-	-	7.090	0.0182
29.18	-	-	7.016	0.0180
30.83	79.28	0.482	-	-
AUXILIARY INFORMATION				
METHOD/APPARATUS/PROCEDURE:		SOURCE AND PURITY OF MATERIALS:		
The method of ref 1 was used. Solutions of suitable composition by weight were placed in a soda-glass flask fitted with a thermometer. The flask was constantly shaken in a slowly-heated water bath, and the temperature of onset of cloudiness was noted. Disappearance of cloudiness was also studied. The determinations were repeated in Pyrex-glass flasks, but no difference was observed.		(1) dried over CaO for a week, refluxed 8 h, fractionated; the whole distilled 118.19-118.35°C at 775.3 mm Hg; middle fraction b.p. 117-71°C (corr.) used d ₄ ²⁵ 0.8055		
		(2) not stated.		
		ESTIMATED ERROR:		
		Temperature: repeated observations within 0.050°C		
		REFERENCES:		
		1. Sidgwick, N.V.; Pickford, P.; Wilsdon, B.H. J. Chem. Soc. <u>1911</u> , 99, 1122.		

COMPONENTS:		ORIGINAL MEASUREMENTS:			
(1) 1-Butanol; C ₄ H ₁₀ O; [71-36-3]		Aoki, Y.; Moriyoshi, T.			
(2) Water; H ₂ O; [7732-18-5]		J. Chem. Thermodyn. <u>1978</u> , 10, 1173-9.			
VARIABLES:		PREPARED BY:			
Temperature: 303-398K		S.C. Valvani; S.H. Yalkowsky; A.F.M.Barton;			
Pressure: 1-2450 atm (0.1-25 MPa)		G.T. Hefter			
EXPERIMENTAL VALUES:					
Mutual solubility of 1-butanol (1) and water (2)					
T/K	p/atm	g(1)/100g sln		α ₁ (compiler)	
		(2)-rich phase	(1)-rich phase	(2)-rich phase	(1)-rich phase
302.95	1	7.0	79.3	0.0180	0.482
302.95	500	9.0	77.7	0.0235	0.459
302.95	970	9.6	76.9	0.0252	0.447
302.95	1150	10.2	76.3	0.0269	0.439
302.95	1240	11.0	76.1	0.0292	0.436
302.95	1500	11.2	75.0	0.0298	0.422
302.95	2000	12.7	74.0	0.0342	0.409
302.95	2250	13.9	72.5	0.0377	0.391
302.95	2450	14.3	72.2	0.0390	0.387
322.75	1	6.9	77.7	0.0177	0.459
322.75	500	8.6	76.0	0.0224	0.435
322.75	1000	9.8	74.4	0.0257	0.414
322.75	1240	10.6	73.5	0.0280	0.403
322.75	1550	11.0	73.2	0.0292	0.399
322.75	2040	12.6	72.2	0.0339	0.387
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AUXILIARY INFORMATION					
METHOD/APPARATUS/PROCEDURE:		SOURCE AND PURITY OF MATERIALS:			
The method,described in ref 1, was that used for previous studies on 2-butanol and iso-butanol. Both components were placed in a cut-off glass syringe of about 20 cm ³ capacity used as a sample vessel, which was placed in a stainless steel pressure vessel mechanically shaken in an oil thermostat bath (± 0.02 K). After stirring for at least 12 h and then standing for another 6 h at the desired temperature and pressure a sample of the upper layer was withdrawn. Subsequently the pressure vessel was moved, the contents allowed to settle, and the lower layer sampled.		(1) "best grade reagent"; dried by refluxing over CaO and then distilling twice; n ²⁵ = 1.3973, d ²⁵ = 0.8062 g cm ⁻³			
Methanol as a mixing agent was added to the samples before refractometric analysis until the mass ratio of sample to methanol became 5.000 ± 0.037.		(2) de-ionized, distilled from alkaline KMnO ₄ and then redistilled; n ²⁵ = 1.3327			
		ESTIMATED ERROR:			
		Solubility: mean error within ± 0.28 g(1)/100g sln			
		REFERENCES:			
		1. Moriyoshi, T.; Kaneshina, S.; Aihara, K.; Yabumoto, K. J. Chem. Thermodyn. <u>1975</u> , 7, 537.			

COMPONENTS:		ORIGINAL MEASUREMENTS:			
(1) 1-Butanol; C ₄ H ₁₀ O; [71-36-3]		Aoki, Y.; Moriyoshi, T.			
(2) Water; H ₂ O; [7732-18-5]		J. Chem. Thermodyn. 1978, 10, 1173-9.			

EXPERIMENTAL VALUES (continued)					
T/K	p/atm	g(1)/100g sln		x ₁ (compiler)	
		(2)-rich phase	(1)-rich phase	(2)-rich phase	(1)-rich phase
322.75	2250	12.7	71.5	0.0342	0.379
322.75	2450	13.2	70.7	0.0356	0.370
332.65	1	7.0	76.1	0.0180	0.436
332.65	500	7.5	74.5	0.0193	0.415
332.65	1000	8.6	73.2	0.0224	0.399
332.65	1500	9.3	71.7	0.0243	0.381
332.65	2010	9.5	70.5	0.0249	0.368
332.65	2180	10.8	70.0	0.0286	0.362
332.65	2250	10.8	69.2	0.0286	0.353
332.65	2400	11.3	69.0	0.0302	0.351
332.65	2450	11.5	69.0	0.0306	0.351
342.65	1	6.7	75.0	0.0171	0.422
342.65	670	7.8	72.5	0.0202	0.391
342.65	1700	10.2	69.5	0.0269	0.357
342.65	2040	10.7	68.5	0.0283	0.346
342.65	2450	11.9	66.6	0.0318	0.327
362.65	1	7.5	69.7	0.0193	0.359
362.65	1000	10.7	65.4	0.0283	0.315
362.65	1040	-	65.4	-	0.315
362.65	1290	11.3	64.9	0.0302	0.310
362.65	1700	13.2	63.5	0.0356	0.297
362.65	2000	14.5	62.6	0.0396	0.289
362.65	2180	15.5	62.3	0.0427	0.287
362.65	2450	16.1	61.4	0.0446	0.278
372.45	1	9.5	66.8	0.0249	0.328
372.45	1000	12.3	62.8	0.0330	0.291
372.45	1290	12.7	62.3	0.0342	0.287
372.45	1570	13.7	60.7	0.0371	0.273
372.45	1800	15.6	60.3	0.0430	0.270
372.45	1970	16.7	58.9	0.0465	0.258
372.45	2450	18.5	55.2	0.0522	0.231
380.05	1	10.6	62.7	0.0280	0.290
380.05	500	11.5	60.9	0.0306	0.274
380.05	1000	13.2	59.0	0.0356	0.259
380.05	1500	15.8	57.4	0.0436	0.247
380.05	2000	16.7	54.5	0.0465	0.226
380.05	2300	20.3	52.3	0.0583	0.210
380.05	2450	25.5	44.9	0.0768	0.165
382.95	1	11.0	61.7	0.0292	0.281
382.95	500	11.9	60.1	0.0318	0.268

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(2) Water; H ₂ O; [7732-18-5]		J. Chem. Thermodyn. <u>1978</u> , <i>10</i> , 1173-9.			
EXPERIMENTAL VALUES (continued)					
T/K	p/atm	g(1)/100g sln		x ₁ (compiler)	
		(2)-rich phase	(1)-rich phase	(2)-rich phase	(1)-rich phase
382.95	1000	13.9	58.3	0.0377	0.254
382.95	1500	16.5	56.4	0.0458	0.239
382.95	1570	17.2	56.3	0.0481	0.238
382.95	1700	17.3	55.8	0.0484	0.235
382.95	1900	18.0	54.1	0.0506	0.223
382.95	2000	19.6	52.5	0.0559	0.212
382.95	2100	23.1	48.9	0.0680	0.189
382.95	2200	31.6	41.0	0.1009	0.145
384.45	1700	19.4	54.6	0.0552	0.227
384.45	1800	20.0	52.4	0.0573	0.211
384.45	1900	22.5	49.9	0.0659	0.195
384.45	2000	30.8	41.0	0.0976	0.227
386.15	1500	20.2	53.5	0.0579	0.218
386.15	1570	22.2	51.8	0.0649	0.207
386.15	1670	27.5	46.0	0.0844	0.172
387.15	1500	26.4	46.1	0.0802	0.172
388.15	1000	17.8	53.2	0.0500	0.216
388.15	1090	18.0	53.0	0.0506	0.215
388.15	1200	19.0	51.8	0.0539	0.207
388.15	1360	25.9	45.5	0.0783	0.169
388.15	1400	31.2	40.2	0.0992	0.141
390.35	800	18.0	52.9	0.0506	0.214
390.35	900	19.0	50.1	0.0539	0.196
390.35	1000	25.2	45.7	0.0757	0.170
391.25	1	14.5	55.6	0.0396	0.233
391.25	300	15.6	54.6	0.0430	0.227
391.25	500	16.7	53.7	0.0465	0.220
391.25	600	18.0	53.5	0.0506	0.218
391.25	720	19.7	51.8	0.0562	0.207
391.25	770	21.6	48.5	0.0627	0.187
391.25	790	23.5	44.1	0.0695	0.160
392.45	450	25.5	45.5	0.0768	0.169
392.45	500	30.0	39.8	0.0944	0.139
392.95	1	17.2	53.7	0.0481	0.220
392.95	270	21.2	49.7	0.0613	0.194
392.95	340	26.3	46.3	0.0798	0.173
392.95	350	27.1	44.7	0.0829	0.164
395.25	1	18.2	47.7	0.0513	0.182
395.25	150	25.1	41.9	0.0753	0.149
395.25	200	32.3	36.8	0.1039	0.124
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COMPONENTS:

(1) 1-Butanol; $C_4H_{10}O$; [71-36-3](2) Water; H_2O ; [7732-18-5]

ORIGINAL MEASUREMENTS:

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EXPERIMENTAL VALUES (continued)

T/K	p/atm	$g(1)/100g\ sln$		$x_1(\text{compiler})$	
		(2)-rich phase	(1)-rich phase	(2)-rich phase	(1)-rich phase
395.85	1	18.9	48.0	0.0536	0.183
397.05	1	21.8	44.9	0.0635	0.165
397.45	1	24.7	41.8	0.0739	0.148
397.75	1	27.9	38.8	0.0859	0.134

Properties of the critical solutions

p_c/atm	$T_c(\text{UCST})/K$	x_{1c}
1	397.85	0.110
204	395.25	0.113
380	392.95	0.117
500	392.45	0.115
820	391.25	0.118
1000	390.65	0.118
1030	390.35	0.118
1390	388.15	0.118
1500	387.65	0.120
1710	386.15	0.122
2000	384.35	0.122
2010	384.45	0.121
2200	382.95	0.121
2450	381.15	0.123
2500	380.05	0.125

$$(dT_c/dp) = -(12.0 \pm 0.5) \times 10^{-3} \text{ K atm}^{-1} \text{ at } p < 400 \text{ atm}$$

$$-(7.0 \pm 0.7) \times 10^{-3} \text{ K atm}^{-1} \text{ at } 800 < p < 2500 \text{ atm}$$

$$(dx_{1,c}/dT) = -(4.0 \pm 0.5) \times 10^{-4} \text{ K}^{-1}$$